

AI ASSISTANT CODING

ASSIGNMENT-4

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Batch:24

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Q1. Zero-Shot Prompting (Basic Lab Task)

Task:

Write a Python function that classifies a given text as Spam or Not Spam using zero-shot prompting.

Steps:

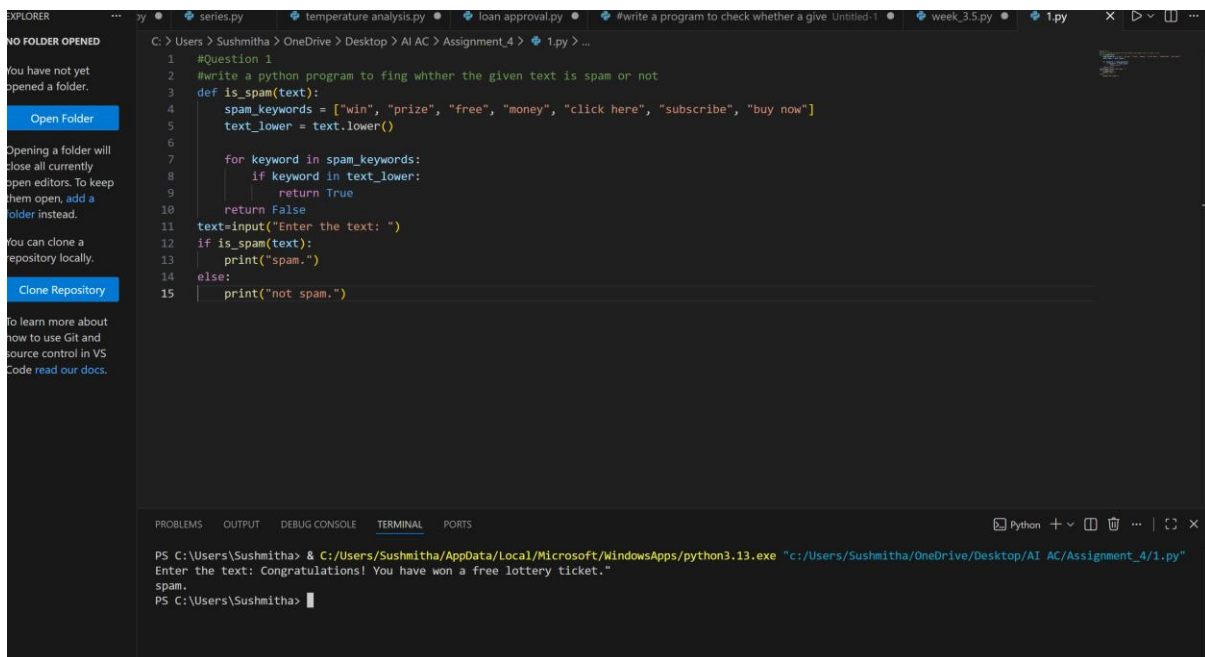
1. Construct a prompt without any examples.
2. Clearly specify the output labels.
3. Display only the predicted label.

Input:

"Congratulations! You have won a free lottery ticket."

Expected Output:

Spam



The screenshot shows a Visual Studio Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a folder named 'Assignment_4' with a file '1.py'. The code editor displays the following Python code:

```
1 #Question 1
2 #Write a python program to find whether the given text is spam or not
3 def is_spam(text):
4     spam_keywords = ["win", "prize", "free", "money", "click here", "subscribe", "buy now"]
5     text_lower = text.lower()
6
7     for keyword in spam_keywords:
8         if keyword in text_lower:
9             return True
10    return False
11    text=input("Enter the text: ")
12    if is_spam(text):
13        print("spam.")
14    else:
15        print("not spam.")
```

The terminal at the bottom shows the command to run the script and the output:

```
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/1.py"
Enter the text: Congratulations! You have won a free lottery ticket.
spam.
PS C:\Users\Sushmitha>
```

Q2. One-Shot Prompting (Emotion detection)

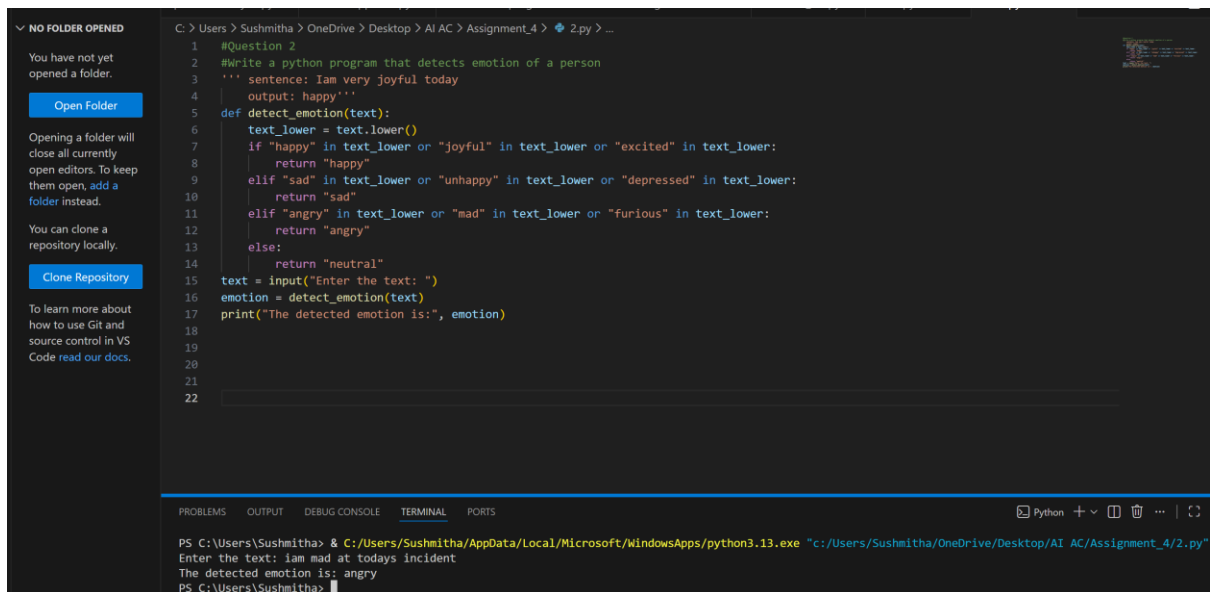
Task:

Write a Python program that detects the emotion of a sentence using one-shot prompting.

Emotions: ['happy', 'sad', 'angry', 'excited', 'nervous', 'neutral']

Steps:

1. Provide one labeled example inside the prompt.
2. Take a sentence as input.
3. Print the predicted emotion



```
1 #Question 2
2 #Write a python program that detects emotion of a person
3 ''' sentence: Iam very joyful today
4 output: happy'''
5 def detect_emotion(text):
6     text_lower = text.lower()
7     if "happy" in text_lower or "joyful" in text_lower or "excited" in text_lower:
8         return "happy"
9     elif "sad" in text_lower or "unhappy" in text_lower or "depressed" in text_lower:
10        return "sad"
11    elif "angry" in text_lower or "mad" in text_lower or "furious" in text_lower:
12        return "angry"
13    else:
14        return "neutral"
15    text = input("Enter the text: ")
16    emotion = detect_emotion(text)
17    print("The detected emotion is:", emotion)
18
19
20
21
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/2.py"

Enter the text: iam mad at todays incident

The detected emotion is: angry

PS C:\Users\Sushmitha>

Q3. Few-Shot Prompting (Student Grading Based on Marks)

Task:

Write a Python program that predicts a student's grade based on marks using few-shot prompting.

Grades:

['A', 'B', 'C', 'D', 'F']

Grading Criteria (to be inferred from examples):

- 90–100 → A
- 80–89 → B
- 70–79 → C
- 60–69 → D
- Below 60 → F

```
C:\Users\Sushmitha> OneDrive > Desktop > AI AC > Assignment_4 > 3.py > ...
1 #Question 3
2 ''' marks-90-100
3     display "a"
4     marks-80-89
5     display "b"
6     marks-70-79
7     display "c"
8     marks-60-69
9     display "d"
10    marks-below 60
11    display "f" '''
12 def students_grade(marks):
13     if 90 <= marks <= 100:
14         return "a"
15     elif 80 <= marks < 90:
16         return "b"
17     elif 70 <= marks < 80:
18         return "c"
19     elif 60 <= marks < 70:
20         return "d"
21     elif marks < 60:
22         return "f"
23 marks = int(input("Enter the marks: "))
24 print("The grade is:", students_grade(marks))
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/3.py"
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/3.py"
Enter the marks: 67
The grade is: d
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/3.py"
Enter the marks: 90
The grade is: a
PS C:\Users\Sushmitha>
```

Q4. Multi-Shot Prompting (Indian Zodiac Sign Prediction using Month Name)

Task:

Write a Python program that predicts a person's Indian Zodiac sign (Rashi) based on the month of birth (month name) using multi-shot prompting.

Indian Zodiac Order (Simplified Month-Based Model): The Indian Zodiac cycle starts in March with Mesha and follows this order:

March → Mesha

April → Vrishabha

May → Mithuna

June → Karka

July → Simha

August → Kanya

September → Tula

October → Vrischika

November → Dhanu

December → Makara

January → Kumbha

February → Meena

```
C:\Users\Sushmitha> OneDrive\Desktop> AI AC> Assignment_4> 4.py > ...
1 #Question 4
2 """ March → Mesha
3     April → Vrishabha
4     May → Mithuna
5     June → Karka
6     July → Simha
7     August → Kanya
8     September → Tula
9     October → Vrischika
10    November → Dhanu
11    December → Makara
12    January → Kumbha
13    February → Meena"""
14 def month_to_zodiac(month):
15     month = month.lower()
16     zodiac_signs = {
17         "march": "Mesha",
18         "april": "Vrishabha",
19         "may": "Mithuna",
20         "june": "Karka",
21         "july": "Simha",
22         "august": "Kanya",
23         "september": "Tula",
24         "october": "Vrischika",
25         "november": "Dhanu",
26         "december": "Makara",
27         "january": "Kumbha",
28         "february": "Meena"
29     }
30     return zodiac_signs.get(month, "Invalid month")
31 month=input("Enter the month: ")
32 print("The zodiac sign is:", month_to_zodiac(month))
33

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] [ ]

PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/4.py"
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/4.py"
Enter the month: january
Enter the month: january
The zodiac sign is: Kumbha
PS C:\Users\Sushmitha> |
```

Q5. Result Analysis Based on Marks

Task: Write a Python program that determines whether a student

Passes or Fails based on marks using Chain-of-Thought (CoT)

prompting.

Result Categories:

['Pass', 'Fail']

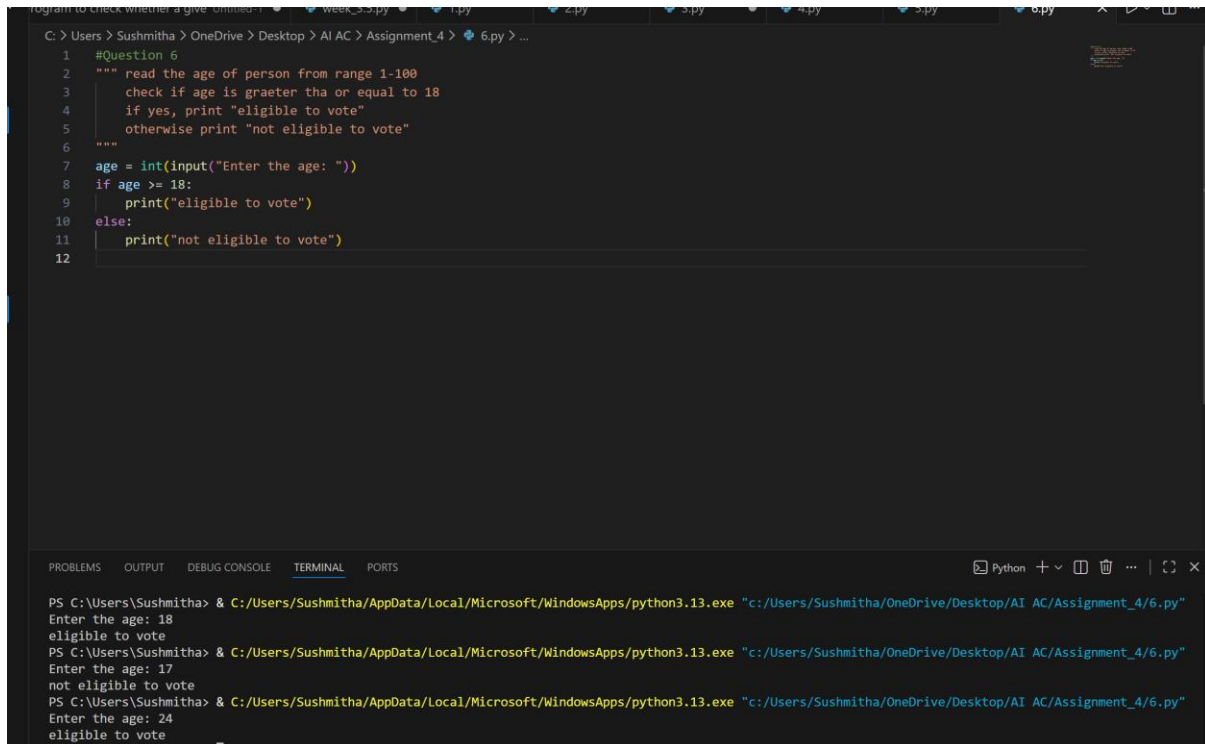
```
C:\Users\Sushmitha> OneDrive\Desktop> AI AC> Assignment_4> 5.py > check_pass_fail
1 #Question 5
2 """ read marks of students from range 0-100
3     check if marks are greater than 40
4     if yes display "pass"
5     if no display "fail" """
6 def check_pass_fail(marks):
7     if 0 <= marks <= 100:
8         if marks >= 40:
9             return "pass"
10        else:
11            return "fail"
12    else:
13        return "invalid marks"
14 marks = int(input("Enter the marks: "))
15 print(check_pass_fail(marks))
16
17
18
19

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] [ ]

PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/5.py"
Enter the marks: 67
pass
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/5.py"
Enter the marks: 40
pass
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/5.py"
Enter the marks: 39
fail
PS C:\Users\Sushmitha> |
```

Q6 Voting Eligibility Check (Chain-of-Thought Prompting)

Task: Write a Python program that determines whether a person is eligible to vote using Chain-of-Thought (CoT) prompting.

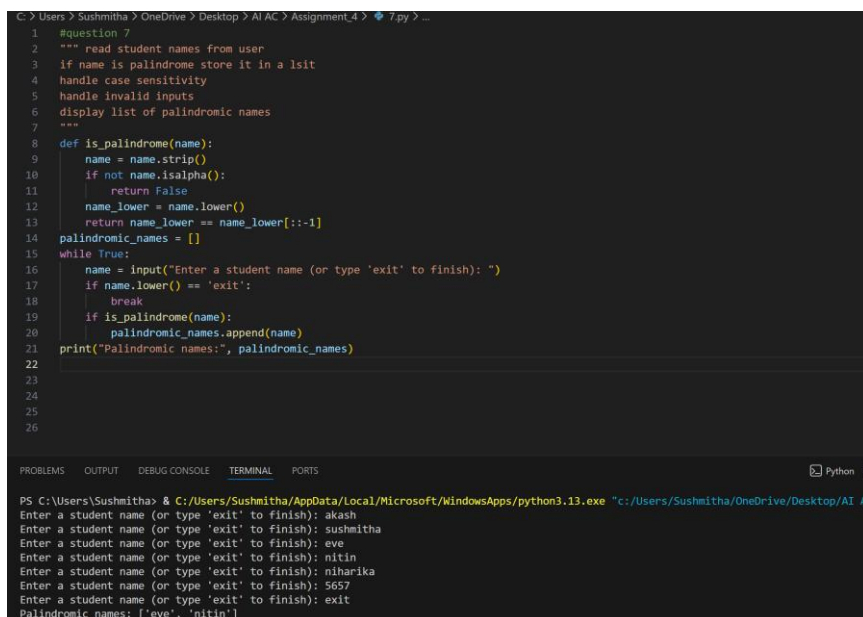


```
C:\Users\Sushmitha> OneDrive > Desktop > AI AC > Assignment_4 > 6.py > ...  
1 #Question 6  
2 """ read the age of person from range 1-100  
3 check if age is graeter tha or equal to 18  
4 if yes, print "eligible to vote"  
5 otherwise print "not eligible to vote"  
6 """  
7 age = int(input("Enter the age: "))  
8 if age >= 18:  
9     print("eligible to vote")  
10 else:  
11     print("not eligible to vote")  
12
```

```
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/6.py"  
Enter the age: 18  
eligible to vote  
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/6.py"  
Enter the age: 17  
not eligible to vote  
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/6.py"  
Enter the age: 24  
eligible to vote
```

Q7 Prompt Chaining (String Processing – Palindrome Names)

Task: Write a Python program that uses the prompt chaining technique to identify palindrome names from a list of student names.



```
C:\Users\Sushmitha> OneDrive > Desktop > AI AC > Assignment_4 > 7.py > ...  
1 #question 7  
2 """ read student names from user  
3 if name is palindrome store it in a list  
4 handle case sensitivity  
5 handle invalid inputs  
6 display list of palindromic names  
7 """  
8 def is_palindrome(name):  
9     name = name.strip()  
10    if not name.isalpha():  
11        return False  
12    name_lower = name.lower()  
13    return name_lower == name_lower[::-1]  
14 palindromic_names = []  
15 while True:  
16     name = input("Enter a student name (or type 'exit' to finish): ")  
17     if name.lower() == 'exit':  
18         break  
19     if is_palindrome(name):  
20         palindromic_names.append(name)  
21 print("Palindromic names:", palindromic_names)  
22  
23  
24  
25  
26
```

```
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Sushmitha/OneDrive/Desktop/AI AC/Assignment_4/7.py"  
Enter a student name (or type 'exit' to finish): akash  
Enter a student name (or type 'exit' to finish): sushmitha  
Enter a student name (or type 'exit' to finish): eve  
Enter a student name (or type 'exit' to finish): nitin  
Enter a student name (or type 'exit' to finish): niharika  
Enter a student name (or type 'exit' to finish): 5657  
Enter a student name (or type 'exit' to finish): exit  
Palindromic names: ['eve', 'nitin']
```

Analysis)

```
C:\Users\Sushmitha> OneDrive\ Desktop > AI AC > Assignment_4 > 8.py > ...

1 #Question 8
2 """ read words from user
3     count the length of each and store it in a variable
4     if variable <5 display as short
5     otherwise display as long
6     display the result of each word"""
7 def classify_word_length(word):
8     length = len(word)
9     if length < 5:
10         return "short"
11     else:
12         return "long"
13 words = input("Enter words separated by spaces: ").split()
14 for word in words:
15     classification = classify_word_length(word)
16     print(f"The word '{word}' is {classification}.")
17
18
19
```